An evo-devo neurocomputational model of the role of DCX in neuronal migration and neocortical lamination

The origin of the neocortex has been proposed to be linked to olfactory and navigational adaptations, and the expression of doublecortin (DCX) in the superficial layers may reflect this evolutionary convergence. From an evo-devo perspective, we asked how loss of DCX function affects neuronal migration and cortical lamination. Our hypothesis was that reduced DCX disrupts radial migration, preventing neurons from reaching their correct laminar positions, particularly in superficial layers. We developed an agent-based model in NetLogo that integrates neuronal migration and layer formation through apoptosis. DCX was implemented as a modulator of migratory properties, and three levels of loss of function (20%, 40%, and 60%) were simulated, with three replicates per condition. The simulations reproduced a typical sequence of radial migration, in which neurons originate in the intermediate zone, first reach the deep layer (Layer 6), and subsequently the superficial layer (Layer 2). The effect of DCX disruption depended on the position along the migratory trajectory analyzed. In Layer 6, intermediate and high levels of loss of function produced the strongest reductions in migration, with the most severe condition showing an approximately tenfold decrease relative to control. In Layer 2, migration also decreased as DCX loss of function increased, with an approximately fivefold reduction in the most severe condition. These results suggest that DCX may have contributed to critical functions in radial migration during the evolution of gyrencephalic brains.

Keywords: neuronal migration, agent-based modeling, neocortex, doublecortin, computational modeling, evolution.